Digital Literacy and Mobile Technology Use in Homework: A Study of Secondary Students in Zomba District, Malawi

Abstract

This study investigates the extent of mobile technology use for homework and associated digital literacy levels among secondary school students in Zomba District, Malawi. A mixed-methods approach was employed across six schools, involving 460 students and 18 teachers. Findings indicate that 277 students (60.0%) rely on smartphones for approximately 80% of their academic tasks. Urban students lead in usage (143), followed by semi-urban (80) and rural (54). The remaining 184 lack consistent access, often borrowing devices. Common applications include ChatGPT, Meta AI, Gemini AI, YouTube for tutorial videos, WhatsApp groups for sharing past papers and attending part-time lessons and Google searches for PDFs. While urban students enjoy full device ownership and continuous data plans, rural users face high costs and depend on discounted TNM "Yanga" night bundles when connectivity improves. Despite widespread student use, most teachers prohibit phones, citing misuse on social platforms—confirmed by 78% of students who admit non-academic activity. Learners demonstrate strong operational skills but limited ability to assess online content critically. Strategic recommendations include zero-rating educational websites, delivering targeted teacher training in mobile learning, and establishing district-level mobile use policies.

Keywords: Digital literacy, mobile learning, homework, secondary education, Malawi, Zomba, ICT4E

1. **Introduction**

Malawi's National Education Sector Investment Plan (2020–2030) prioritizes ICT integration to enhance learning outcomes. However, computer availability remains low, with mobile phones serving as the primary internet gateway. In Zomba District, where infrastructure constraints affect rural and peri-urban schools, students increasingly turn to smartphones to complete assignments.

This research addresses three critical gaps:

1. Limited local evidence on how students apply mobile technology to academic work

2. Insufficient understanding of the link between mobile use and digital literacy development

3. Absence of documented teacher perspectives on mobile-based homework practices

Research questions:

RQ1: To what extent do secondary students in Zomba use mobile devices for homework?

RQ2: What digital literacy capabilities do they exhibit?

RQ3: What factors hinder effective and equitable use?

2. **Literature Review**

Across Africa, mobile phones are the dominant digital access point for learners. The UNFPA TEENS initiative in Balaka demonstrated that tablet access combined with mentorship improved girls' engagement and language skills. A Zambian survey of 281 educators revealed moderate-to-high digital proficiency but inconsistent classroom application due to cost and curricular misalignment. A systematic review on ICT integration in Tanzania emphasized that success depends on sustained skill development, curriculum coherence, and user motivation. Globally, mobile learning enhances collaboration and comprehension, yet raises concerns about information credibility and access equity. This study builds on existing insights by examining these dynamics within Zomba's unique educational landscape.

3. **Methodology**

3.1 Design: Cross-sectional descriptive study using mixed methods, conducted from September to November 2026.

3.2 Setting: Six public secondary schools stratified by location—two urban, two peri-urban, two rural.

3.3 Participants: 460 Form 2–4 students selected via stratified random sampling; 18 teachers through purposive sampling.

3.4 Instruments:

- Student questionnaire: Assessed device access, usage patterns, self-rated digital skills, and challenges. Adapted from UNESCO's Digital Literacy Global Framework.

- Teacher interviews: Semi-structured discussions on attitudes, homework expectations, and observed student behaviors.

- Practical skills assessment: Five tasks—performing a Google search, downloading a PDF, sharing a file via WhatsApp, evaluating real vs. fake news sites, and creating a basic document on a phone.

3.5 Analysis: Quantitative data analyzed using SPSS v29 (descriptive statistics, chi-square tests for urban-rural comparisons). Qualitative data coded thematically using NVivo.

3.6 Ethics: Approved by the Zomba District Education Office. Informed consent and assent obtained. All participant identities anonymized.

4. **Results**

4.1 Access and Usage Patterns

Among 460 students, 277 (about 60.0%) reported using smartphones for roughly 80% of their homework. Urban students accounted for 143 users, semi-urban for 80, and rural for 54. The remaining 184 had no personal access and relied on borrowed devices. Key tools included ChatGPT, Meta AI, YouTube for instructional videos, WhatsApp groups for collaborative learning, and Google for locating academic materials. Device ownership was universal in urban areas (100%), lower in semi-urban zones, and lowest in rural communities. Data affordability emerged as a key constraint: while urban students maintained active bundles, rural peers depended on TNM's discounted "Yanga" night packages to access faster networks at reduced rates.

Table 1

Location	Number of students using smartphones for 80% of homework	% of total sample
Urban	143	31.1
Semi-urban	80	17.4
Rural	54	11.7
Total	277	60.2

Number = 460

4.2 Digital Literacy Assessment

Students displayed strong functional competencies but weak critical evaluation skills. Performance across key areas:

Table 2

Skill	Number of students	% proficient
Basic navigation	422	91.7
Downloading and saving files	286	62.1
Creating or sharing documents	222	48.3
Evaluating source credibility	123	26.7
Privacy setting management	88	19.2

Number = 460

From this data, girls were very weak at this as compared to boys

4.3 Teacher Perspectives

Most teachers enforce strict phone bans, believing students prioritize TikTok, Facebook, and messaging over academics—a concern echoed by 78% of students who acknowledged non-educational use. Only three teachers had participated in ICT-related professional development in the past two years. Primary concerns included distraction, academic dishonesty, inequity, and absence of institutional policy.

4.4 Key Barriers

1. High data costs—reported by 63%, particularly acute in rural areas

2. Device sharing and household conflicts—51%, more prevalent outside urban centers

3. Unreliable network coverage—47% in rural vs. 12% in urban; many rural students work late at night for better speeds

4. Lack of instructional guidance—58% stated no one taught them how to use phones effectively for learning

5. Institutional restrictions—phone confiscation drives covert usage

5. Discussion

Students in Zomba are already leveraging mobile technology for learning, though access and support remain uneven. Urban learners benefit from consistent connectivity and ownership, while rural students strategically optimize limited, low-cost access periods. These patterns mirror findings from Balaka's TEENS project, where structured access and mentorship enhanced outcomes. The high rate of non-academic use confirms teacher concerns, yet outright bans overlook the reality that 60% of students depend on phones for the majority of their academic work.

As seen in Zambia and Tanzania, sustainable integration hinges on teacher capacity and curricular alignment. In Zomba, the challenge is no longer mere access—it is a gap in skills and institutional support.

Policy implication: Partnering with Malawi Schoolnet to zero-rate key educational domains would significantly reduce cost barriers for rural users.

Sector opportunity: Teacher training institutions should introduce "mobile pedagogy" modules. Industry stakeholders can develop affordable devices and offline-first educational apps compatible with 2G/3G networks.

6. **Conclusion & Recommendations**

Mobile technology is now integral to homework completion in Zomba's secondary schools. The prevailing digital divide reflects disparities in digital competence and support, not just connectivity.

Recommendations:

1. Ministry of Education & Mobile Network Operators: Implement zero-rating for educational platforms—including Wikipedia, Khan Academy, MANEB, and national past paper portals.

2. Zomba District Education Office: Issue clear mobile-use guidelines and deliver CPD programs focused on mobile learning strategies and digital citizenship.

3. Schools: Replace blanket bans with balanced policies—permit phones for research, assign mobile-based tasks, and establish norms for academic WhatsApp groups.

4. NGOs and Private Sector: Scale proven models like Balaka's Safe Space and tablet mentorship, with emphasis on rural female learners.

5. Future Research: Conduct longitudinal studies to assess the impact of mobile-assisted homework on MSCE examination results.

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